

Churn Prediction & Sentiment Agent

Proposed by: CS Studio

Hackathon scope: Dedicated agent for one client company

1. Idea Summary

The Churn Prediction & Sentiment Agent is a dedicated Customer Support agent for one client company. It brings together customer signals, explains the health of the client relationship, identifies churn risk early, and suggests practical improvement actions.

The agent is not a multi-client platform. Each deployment is configured for one company and uses only that company's context and data.

2. The Problem

Client frustration can grow quietly across emails, chats, tickets, slow responses, unresolved issues, and lower product usage. Customer Support teams may see these signals separately, but they do not always see the full picture early enough.

This can lead to late action, dissatisfied stakeholders, and account loss.

3. Users and Scope

The main users are Customer Support and Customer Success teams.

The agent focuses on one client company. It understands the people, priorities, product areas, history, and communication style of that client. This allows it to judge whether a signal is normal, important, or critical for that specific relationship.

4. Client Context and Knowledge Base

Before use, the team provides a client profile. This can include:

- Business goals and expected outcomes
- Key contacts and roles, such as CTO, sponsor, or developer
- Critical products, features, and workflows
- Communication preferences and tone
- Service agreements, open commitments, and known risks
- History of incidents, escalations, and improvement plans

This context helps the agent understand severity. For example, a negative message from a CTO about a critical feature may require more attention than a similar message about a minor topic.

5. Data Sources

The agent can combine:

- Support tickets and ticket status
- Email and chat conversations
- Tone and sentiment in communication
- Response and resolution times
- Repeated issues and escalations
- Product usage and changes in activity
- Customer satisfaction scores, such as CSAT or NPS, when available

6. How the Agent Works

1. It collects relevant signals from the available data sources.
1. It compares each signal with the client profile and current commitments.
1. It identifies positive and negative patterns.
1. It calculates a **Churn Risk Score** and a client health status: Healthy, Watch, or At Risk.
1. It explains the main reasons behind the score with evidence.
1. It proposes a prioritized improvement plan for the human team to review.

The score is a risk estimate, not a guaranteed prediction of cancellation.

7. Outputs for the Team

The agent provides:

- Overall client health status
- Churn Risk Score and recent trend
- Positive and negative signals
- Key risks, with supporting evidence
- Priority actions and suggested owners
- Draft communication or outreach messages
- A clear improvement plan

The human team validates recommendations before taking action or sending any message.

8. Hackathon MVP

The MVP should use simulated data for one client company:

- A client profile with key stakeholders and business context
- Sample emails, chats, tickets, and product activity data
- A simple health dashboard
- Churn Risk Score with clear reasons
- A prioritized list of risks
- An improvement plan and one draft outreach message

9. Optional Feature: Feedback Memory

If time allows, the team can add a feedback step. A user can mark an alert as correct, a false alarm, important, or resolved. The agent then saves that feedback to improve future recommendations for the same client.

This feature is optional. The main MVP works without automatic learning.

10. Demo Success Criteria

A strong demo should make three things clear:

1. What is going wrong with the client relationship.
1. Why it matters, based on the client context and evidence.
1. What the Customer Support team should do next.

11. Boundaries

- One deployment serves one client company.
- The agent does not automatically send emails or make decisions.
- Human review is required for recommendations and communications.
- The first version uses a transparent risk score based on available signals.